Land Rental Market Participation for Sub-Saharan Africa

Reproducible workflow for survey-derived statistics on land rental market participation

2026-06-20

Contents

1. Overview

This document present summary statistics on rural land rental market participation (and related variables) derived from nationally representative rural household survey datasets, primarily taken from the World Bank's LSMS-ISA project. the objective is to make the assembly of these variables completely transparent and reproducible from source files. at present, all source files used are taken from the LSMS-ISA project, with the exception of the Rural Agricultural Livelihoods Survey (RALS) data for Zambia, which is used by permission from the Indaba Agricultural Research Institute (IAPRI). The replication code and outputs are provided in the GitHub page where this document is found. The source files need to be downloaded (using links to LSMS-ISA data provided in section 7; RALS data may be accessed through permission from IAPRI) and made available to scripts to enable the workflow to run on a users computer.

2. Coverage

At present, eight nationally representative farm-household surveys across Sub-Saharan Africa, seven from the World Bank LSMS-ISA program and one (Zambia RALS) from IAPRI-MSU.

Table 1. Survey coverage - countries, source surveys, survey years, and the spatial unit. The reporting unit is the *parcel*: in some surveys it is subdivided into fields/plots, in others it is a single land unit.

Country	Survey (source)	Survey years	Spatial unit (reporting unit)
Ethiopia	ESS (LSMS-ISA)	2012, 2014, 2016, 2019, 2022	parcel (subdivided into fields)
Malawi	IHPS (LSMS-ISA)	2010, 2013, 2016, 2019	garden (subdivided into plots)
Mali	EACI (LSMS-ISA)	2014, 2017	parcelle (single land unit)
Niger	ECVMA (LSMS-ISA)	2011, 2014	parcelle (single land unit)
Nigeria	GHS-Panel (LSMS-ISA)	2011, 2013, 2016, 2019, 2023	plot (single land unit)
Tanzania	NPS (LSMS-ISA)	2009, 2011, 2013, 2015, 2019	plot (single land unit)
Uganda	UNPS (LSMS-ISA)	2009, 2010, 2011, 2013, 2015, 2018, 2019	parcel (subdivided into plots); season 1 reported
Zambia	RALS (IAPRI-MSU)	2012, 2015, 2019	field (single land unit)
Tanzania (ASC)	Agric. Sample Census (NBS)	2009, 2019	household (land by tenure category)

3. Definitions and methods

The unit of analysis is the the tenure-bearing land unit in each survey; this is the *field* in single-level surveys and the *parcel* above fields/plots in Ethiopia, Uganda and Malawi. For convenience, we refer to all of these units assembed for the pooled analysis as *parcels*.

For every parcel we record:

Table 2. Output variables - the variables constructed for each unit.

Output variable	Definition	Type
`parcel_rentedin`	Parcel rented or sharecropped IN	0/1
`parcel_rentedout`	Parcel rented or sharecropped OUT	0/1
`parcel_certificate`	Parcel has a land certificate / document	0/1
`parcel_purchased`	Parcel acquired through purchase	0/1 (`.` if not asked)

Output variable	Definition	Type
`parcel_area_ha`	Cultivated parcel area (Σ field GPS, else self-reported)	hectares
`n_fields`	Number of cultivated fields on the parcel	count
`season`	Cropping season (1 = single-season country / Uganda season 1; 2 = Uganda season 2)	1/2
`weight`	Household survey weight	analytic/probability
`ea_id`	Enumeration area (survey PSU)	id
`strataid`	Survey design stratum	id
`country` `wave` `year` `hh_id` `holder_id` `parcel_id`	Identifiers	—

Area cleaning. `parcel_area_ha` is **top-coded**: values above `${area_max}` (40 ha, set in `MASTER.do`) are treated as data-entry errors and set missing. Smallholder parcels rarely exceed this, but the raw self-reported areas carry unit-entry outliers (e.g. tens of thousands of "acres") that otherwise inflate the *mean* (the median is unaffected). This matters most where a parcel falls back to self-reported area because GPS is missing; GPS-measured areas are well-behaved.

Weighting and inference. All shares are survey-weighted using each round's household weight, with the survey design set as `svyset psu [pw=weight], strata(strata)` where the PSU and stratum are made unique by country x wave; single-PSU strata are centered. Confidence intervals are design-based (Taylor linearization). Tables report season 1 (the only season for all countries except Uganda, which is computed for both and reported for season 1).

Please note that the pooled data draws on survey rounds, some of which are panel data (repeated visits to the same households) and new cross-sectional survey rounds (where households are not repeated from prior rounds). The statistics reported here are repeated cross-sectional, design-weighted prevalence estimates rather than panel or transition estimates. For each survey round we apply that round's official household weight, which the data producers construct to make the round representative of its target population and which already incorporates their own unit non-response and, for the panel surveys, attrition adjustments. Because each estimate targets the population at the time of that round rather than a fixed cohort followed across waves, between-wave attrition does not bias a given year's share.

As noted above, several of the surveys are independent cross-sections rather than panels - Mali's EACI and Niger's ECVMA each draw fresh samples - so a common attrition correction would be neither feasible nor meaningful across the eight surveys. We therefore read the over-time figures as a sequence of representative snapshots rather than as within-household trajectories. The lone exception is Zambia's 2019 RALS round, for which only a panel weight is released; that estimate represents the followed panel, and is noted as such.

4. Interpretion guidance

Missing by design (read this before interpreting the tables)

Some variables are **missing `(.)` for entire country-years**. These are **structural** missings: the survey simply did not ask the relevant question that round, so the concept is *unmeasured*, not zero and not item non-response. A `.` here means "not collected this wave," and these country-years must be **excluded** (not treated as 0) when computing rates or trends. This is why the QC estimates one variable at a time - a joint `mean/svy: mean` would casewise-drop these whole country-years from the *other* variables too.

Table 3. Structurally missing variables - items not collected in particular country-years.

Country	Variable	Missing year(s)	Why
Ethiopia	`parcel_purchased`	2012, 2014	ESS11/ESS13 acquisition question had **no "purchased" category**; "Purchased" (code 7) is first offered in wave 3 (2016).
Malawi	`parcel_certificate`	2010	IHPS 2010 has **no title/ownership-document question** for the plot.
Malawi	`parcel_certificate`	2019	The 2019 round **dropped** the title/document question.
Malawi	`parcel_purchased`	2019	The 2019 round **dropped the categorical "how acquired" question** entirely (only "from whom" and "year acquired" remain), so acquisition mode - including purchase - is not identifiable.
Mali	`parcel_rentedout`	2014, 2017	EACI surveys only the parcels a household **operates**, so land rented/lent **out** is out of frame (no rented-out code in 2014; the 2017 "Louee/Pretee" code flags only 5 of ~24,250 parcels).
Nigeria	`parcel_certificate`	2011	GHS wave 1 has **no certificate-of-occupancy question** in the tenure module (added from wave 2).
Tanzania	`parcel_purchased`	2009, 2011, 2013	The early NPS "ownership status" question has **no acquisition-mode / purchase category** (added when the question was redesigned to "how was this plot acquired?" in NPS4, 2015).
Uganda	`parcel_purchased`	2018	UNPS7 (2018/19) fielded a **reduced panel re-interview** roster for owned parcels: the acquisition item `s2aq8` was **not administered** (0 of 4,368 parcels), so acquisition mode - including purchase - is unidentifiable. (Area and tenure-system were also only re-collected for the ~25% of parcels that were new or re-measured; see Uganda §6.)

All other variables are populated in every country-year shown. (Within a populated country-year, ordinary item non-response is handled the usual way - e.g. a parcel with no usable area is missing on `parcel_area_ha` only.)

Rented-in: sharecropping coverage (read before comparing rented-in levels)

`parcel_rentedin` is meant to capture access to land for a payment, which in principle includes **sharecropping** (a share-of-output payment) as well as fixed cash/kind rental. Surveys differ in whether sharecropping is recorded *separately*, so coverage varies:

Table 4. Sharecropping coverage in `rented_in`, by survey.

Country	Sharecropping in `rented_in`?	Basis
Ethiopia	**Yes**	acquisition has both a "rent" and a separate "sharecrop" code
Mali	**Yes**	occupation mode includes *metayage* (sharecropping)
Tanzania	**Yes**	tenure has "rented in" plus a separate "shared-rent" (sharecrop) code
Zambia	**Yes**	field land-use "rented in" is defined as *cash or in-kind*
Malawi	Partial / unclear	"leasehold / rented / tenant" categories; no explicit sharecrop code
Nigeria	Cash rental	"rented" acquisition code; sharecropping not separately identified
Uganda	Cash rental (likely undercounts sharecrop)	rented-in keyed on rent *paid > 0*; in-kind shares may not register
Niger	**No sharecropping at all**	"location" (cash rental) only; the survey has no sharecropping category

Where sharecropping is not separately coded (Niger entirely; Nigeria/Malawi/Uganda weakly), the cross-country rented-in levels are **lower bounds** on total rental-market participation. (Flagged for the questionnaire review.)

Two further round-specific caveats. Uganda's 2018 (UNPS7) round used a reduced panel re-interview: the owned-parcel acquisition item was not administered (so purchase is missing that round) and area/tenure were re-collected for only ~25% of parcels. Zambia's 2019 (RALS) release carries only a panel weight (no cross-sectional weight), so the 2019 endpoint represents the followed panel, not a fresh cross-section.

5. Results

Survey-weighted shares by country and survey year (season 1). A dash (-) marks structurally missing items (question not asked that round; see Section 4).

5.1 Share of households with one or more plot

Table 5. Share of households with at least one plot that is rented-in, rented-out, purchased, or holds a land certificate, by country and survey year (season 1; survey-weighted). A dash (-) denotes an item not collected that round.

Country	Year	Rented-in	Rented-out	Purchased	Has certificate
Ethiopia	2012	0.248	0.073	-	0.486
	2014	0.259	0.132	-	0.537
	2016	0.277	0.178	0.099	0.586
	2019	0.242	0.117	0.068	0.790
	2022	0.234	0.050	0.082	0.782
Malawi	2010	0.121	0.004	0.039	-
	2013	0.117	0.006	0.031	0.026
	2016	0.119	0.024	0.065	0.019
	2019	0.109	0.023	-	-
Mali	2014	0.030	-	0.018	0.075
	2017	0.010	-	0.008	0.063
Niger	2011	0.232	0.001	0.141	0.126
	2014	0.038	0.005	0.149	0.055
Nigeria	2011	0.103	0.002	0.097	-
	2013	0.104	0.006	0.071	0.109
	2016	0.090	0.006	0.088	0.075
	2019	0.187	0.019	0.243	0.174
	2023	0.145	0.013	0.115	0.130
Tanzania	2009	0.112	0.013	-	0.108
	2011	0.075	0.012	-	0.144
	2013	0.061	0.012	-	0.116
	2015	0.093	0.012	0.288	0.146
	2019	0.152	0.029	0.432	0.218
Uganda	2009	0.214	0.030	0.379	0.226
	2010	0.202	0.004	0.350	0.168
	2011	0.136	0.012	0.298	0.159
	2013	0.172	0.011	0.352	0.256
	2015	0.232	0.019	0.366	0.218
	2018	0.210	0.023	-	0.127
	2019	0.188	0.024	0.346	0.075
Zambia	2012	0.030	0.005	0.056	0.086
	2015	0.038	0.011	0.068	0.054
	2019	0.045	0.018	0.079	0.056

5.2 Share of plots

Table 6. Share of plots that are rented-in, rented-out, purchased, or hold a land certificate, by country and survey year (season 1; survey-weighted). A dash (-) denotes an item not collected that round.

Country	Year	Rented-in	Rented-out	Purchased	Has certificate
Ethiopia	2012	0.130	0.032	-	0.458
	2014	0.117	0.053	-	0.510
	2016	0.111	0.064	0.040	0.555
	2019	0.152	0.068	0.033	0.717
	2022	0.134	0.022	0.035	0.688
Malawi	2010	0.082	0.002	0.031	-
	2013	0.082	0.003	0.026	0.025
	2016	0.091	0.015	0.042	0.014
	2019	0.075	0.013	-	-
Mali	2014	0.013	-	0.008	0.041
	2017	0.004	-	0.003	0.031
Niger	2011	0.143	0.000	0.082	0.085
	2014	0.024	0.002	0.098	0.041
Nigeria	2011	0.090	0.002	0.069	-
	2013	0.101	0.004	0.054	0.090
	2016	0.077	0.004	0.062	0.061
	2019	0.112	0.008	0.131	0.095
	2023	0.093	0.008	0.072	0.086
Tanzania	2009	0.062	0.007	-	0.070
	2011	0.042	0.006	-	0.110
	2013	0.038	0.007	-	0.112
	2015	0.063	0.007	0.247	0.143
	2019	0.079	0.014	0.288	0.143
Uganda	2009	0.119	0.013	0.265	0.157
	2010	0.126	0.002	0.242	0.105
	2011	0.085	0.006	0.211	0.101
	2013	0.106	0.005	0.242	0.186
	2015	0.139	0.010	0.253	0.151
	2018	0.128	0.012	-	0.091
	2019	0.116	0.013	0.254	0.045
Zambia	2012	0.013	0.002	0.047	0.070
	2015	0.016	0.003	0.057	0.047
	2019	0.017	0.005	0.068	0.047

5.3 Share of farm area (hectares)

Table 7. Share of farm area (hectares) that is rented-in, rented-out, purchased, or holds a land certificate, by country and survey year (season 1; survey-weighted). A dash (-) denotes an item not collected that round; for rented-out it can also denote that the rented-out parcels carry no measured area (they are uncultivated, so no field area is recorded - e.g. Ethiopia 2019, 2022), making the area share undefined rather than zero.

Country	Year	Rented-in	Rented-out	Purchased	Has certificate
Ethiopia	2012	0.100	0.013	-	0.560
	2014	0.127	0.049	-	0.517
	2016	0.132	0.046	0.034	0.556
	2019	0.186	-	0.018	0.731
	2022	0.192	-	0.027	0.740
Malawi	2010	0.077	0.002	0.037	-
	2013	0.102	0.004	0.039	0.026
	2016	0.077	0.014	0.059	0.014
	2019	0.051	0.013	-	-

Country	Year	Rented-in	Rented-out	Purchased	Has certificate
Mali	2014	0.006	-	0.006	0.045
	2017	0.002	-	0.003	0.020
Niger	2011	0.103	0.000	0.076	0.087
	2014	0.022	0.002	0.094	0.045
Nigeria	2011	0.105	0.001	0.088	-
	2013	0.164	0.003	0.054	0.069
	2016	0.068	0.006	0.069	0.061
	2019	0.127	0.015	0.126	0.108
	2023	0.093	0.010	0.089	0.108
Tanzania	2009	0.039	0.007	-	0.070
	2011	0.026	0.012	-	0.124
	2013	0.026	0.019	-	0.152
	2015	0.058	0.011	0.397	0.183
	2019	0.051	0.030	0.432	0.181
Uganda	2009	0.062	0.022	0.259	0.242
	2010	0.066	0.004	0.237	0.162
	2011	0.048	0.017	0.246	0.152
	2013	0.075	0.008	0.286	0.239
	2015	0.094	0.008	0.296	0.224
	2018	0.134	0.003	-	0.125
	2019	0.082	0.011	0.272	0.046
Zambia	2012	0.011	0.003	0.063	0.083
	2015	0.013	0.004	0.084	0.066
	2019	0.013	0.006	0.083	0.062

We observe different participation rates for the different countries, with the highest rates in Ethiopia (23-27% of households renting in) and Uganda (14-22% of households renting in) and the lowest rates in Zambia (<4% of households renting in) and Mali (<2% of households renting in). This conforms somewhat to rural population densities, although some high population density countries have only moderate rates of rental (e.g., Malawi, where household rates of renting in land are 10-12% for the included waves).

The rates of renting in are systematically higher than rates of renting out. This difference is observed for all countries in all waves, although the ratio of renting in to renting out rates varies considerably, with household rates of renting in exceeding household rates of renting out by as little as 56% (in Ethiopia for 2016) to more than 1000% (most waves in Uganda, Nigeria; the first two waves of Malawi; the first wave of Niger). This speaks to systematic distortions in the probability of reporting renting in versus renting out land. Note that in Mali, data on renting out land is not collected at all.

Household rental participation rates (defined as the share of households in the sample which are renting one or more parcels) are systematically higher than parcel-level statistics, as expected. The magnitude of this difference (which is reflective, in part, of the average number of plots per farm) is lowest in Nigeria (where household rates exceed plot-level rental participation rates by <20%) and highest in Ethiopia and Zambia (where household rental participation rates are often >100% greater than plot-level rental participation rates).

Note that in some countries, the wave-on-wave differences in rental rates is quite high, which may reflect suvey differences more than actual changes in rental market participation. Mali is a good example of this, where rent-to-market participation rates at both the household and plot level are three times higher in the first wave than the second wave. the second wave was a much broader sampling frame than the first wave, however, which means that these averages are reflective of different areas and shouldn't be interpreted as changes over time in the same population

6. Trends over time

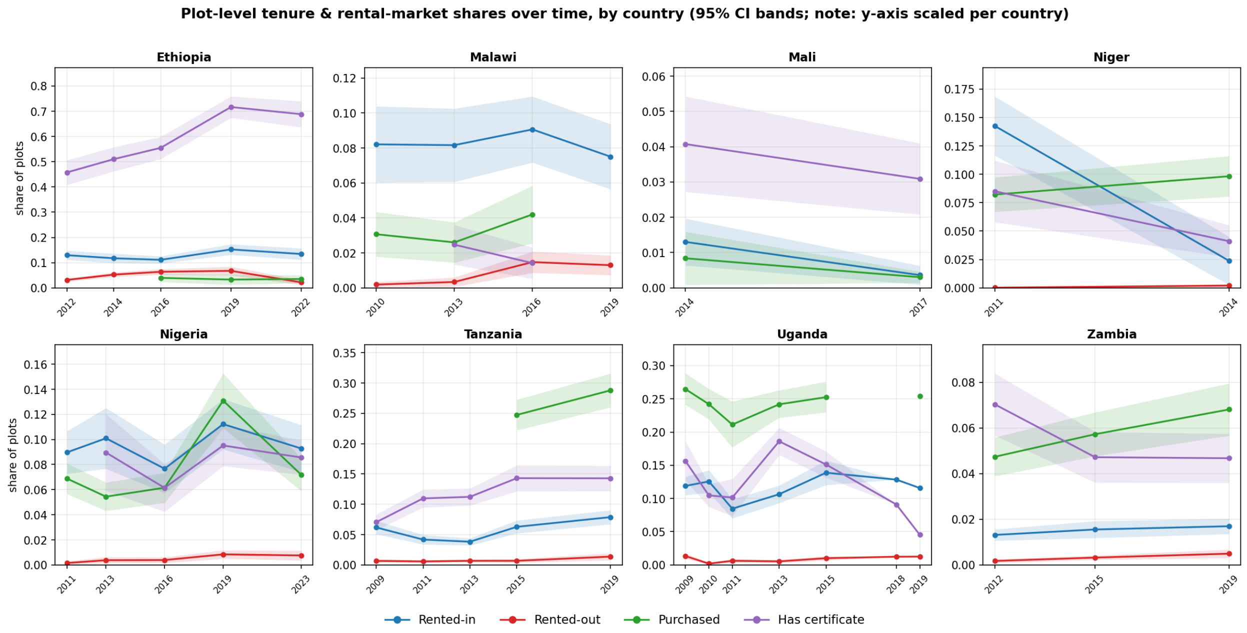

Figure 1: Figure 1. Plot-level tenure and rental-market shares over time, by country, with 95% confidence bands. Y-axis is scaled per country so within-country trends are legible; levels differ across panels.

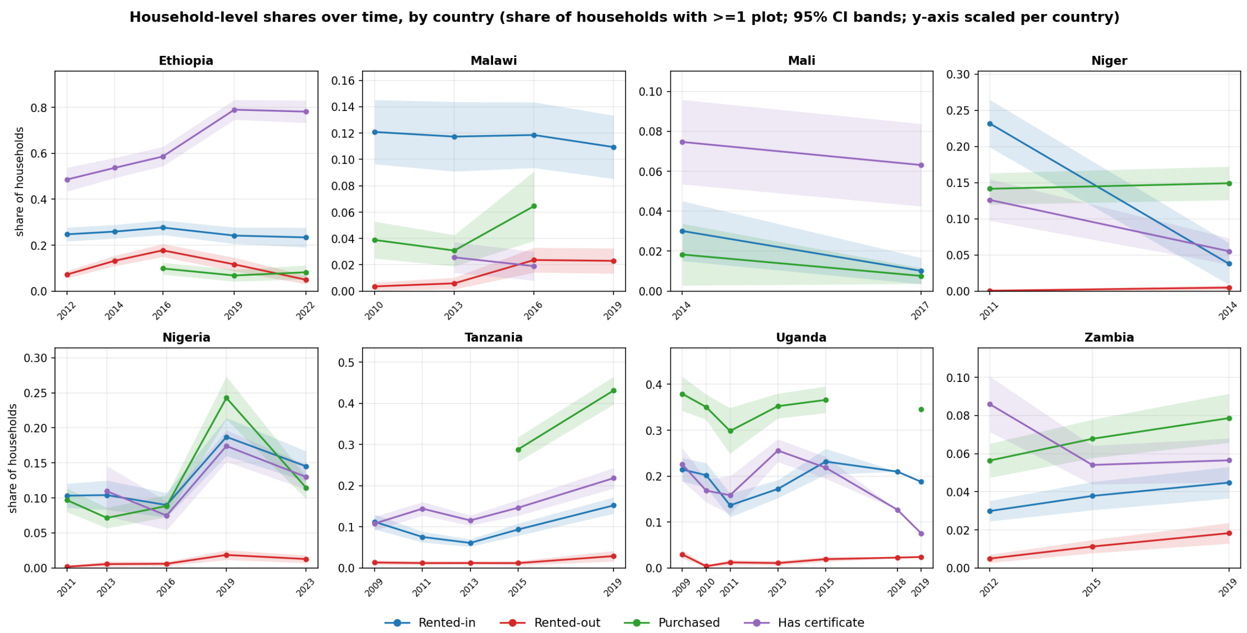

Figure 2: Figure 2. Household-level shares (share of households with at least one plot of each type) over time, by country, with 95% confidence bands. Y-axis is scaled per country, as in Figure 1.

Figures 1 and 2 present these shares over time at the plot and household levels, respectively. With the caveat that waves are not strictly comparable (due to instrument changes, panel attrition, and the sample redesign breaks flagged in Section 4 and the appendix), we may compare levels across time to see if any strong patterns are detectable. For the most part, the patterns are too noisy to ascribe much of a unified story to. Rented land is only increasing consistently over the period observed in Zambia, although at very low base levels. Tanzania shows growth between the first and last waves observed, but not consistently across intervening waves. Ethiopia and Malawi are essentially flat. Altogether, there is not much of a consistent growth story observable in these data.

7. Data sources and reproduction

The repository contains **no microdata**. LSMS-ISA surveys are obtained (free, with registration) from the World Bank Microdata Library; Zambia RALS is obtained from IAPRI. After placing the raw files in the expected folder tree, running `Code/MASTER.do` rebuilds the pooled dataset and `Code/tables_graphs.do` (or `Code/tables_graphs.py`) regenerates every table and figure here. Full per-survey file lists are in `DATA_SOURCES.md`.

All code, reference documentation, and the outputs reproduced here are available in the project's public GitHub repository, https://github.com/chamb244/ssa-land-rental. The repository contains the country-by-country extractors, the pooled-dataset build (`MASTER.do`), the table and figure generators, this report and its build script, and the full variable-provenance reference. Code is released under the MIT License and the accompanying documentation under CC-BY-4.0; the survey microdata themselves are not redistributed and must be obtained from the providers below.

Main source file and catalog link, by survey wave

The table gives, for each survey wave, the primary land/tenure roster file the workflow reads (the complete per-wave file list is in Appendix A). Catalog links are shown where a stable URL is available; entries marked *(pending)* can be located in the World Bank Microdata Library (LSMS-ISA) or, for Zambia, obtained from IAPRI.

Table 8. Primary source file and data catalog, by survey wave.

Country	Year	Main source file	Data catalog
Ethiopia	2012	`sect2_pp_w1.dta`	ESS 2011/12
	2014	`sect2_pp_w2.dta`	ESS 2013/14
	2016	`sect2_pp_w3.dta`	ESS 2015/16
	2019	`sect2_pp_w4.dta`	ESS 2018/19
	2022	`sect2_pp_w5.dta`	ESS 2021/22
Malawi	2010	`ag_mod_d_10.dta`	IHPS 2010/11 (W1)
	2013	`ag_mod_d_13.dta`	IHPS 2013 (W2)
	2016	`ag_mod_b2_16.dta`	IHPS 2016 panel (W3)
	2019	`ag_mod_b2_19.dta`	*(pending: IHPS 2019/20 W4)*
Mali	2014	`EACIEXPLOI_p1.dta`	EACI 2014
	2017	`eaci17_s11bp1.dta`	EACI 2017
Niger	2011	`ecvmaas1_p1.dta`	ECVMA 2011
	2014	`ECVMA2_AS1P1.dta`	ECVMA 2014
Nigeria	2011	`sect11b_plantingw1.dta`	GHS 2010/11
	2013	`sect11b1_plantingw2.dta`	GHS 2012/13
	2016	`sect11b1_plantingw3.dta`	GHS 2015/16
	2019	`sect11b1_plantingw4.dta`	GHS 2018/19
	2023	`sect11b1_plantingw5.dta`	GHS-Panel 2023 (W5)
Tanzania	2009	`SEC_3A.dta`	NPS 2008/09
	2011	`AG_SEC3A.dta`	NPS 2010/11

Country	Year	Main source file	Data catalog
	2013	AG_SEC_3A.dta	NPS 2012/13
	2015	AG_SEC_3A.dta (extended + refresh)	NPS 2014/15
	2019	AG_SEC_3A.dta (extended + refresh)	NPS 2019/20 SDD
Uganda	2009	2009_AGSEC2A.dta	UNPS 2009/10
	2010	AGSEC2A.dta	UNPS 2010/11
	2011	AGSEC2A.dta	UNPS 2011/12
	2013	AGSEC2A.dta	UNPS 2013/14
	2015	AGSEC2A.dta	UNPS 2015/16
	2018	AGSEC2A.dta	UNPS 2018/19
	2019	agsec2a.dta	UNPS 2019/20
Zambia	2012	field.dta	RALS 2012 (IHSN)
	2015	field.dta	RALS 2015 (IHSN)
	2019	field.dta	request from IAPRI (info@iapri.org.zm)
Tanzania (ASC)	2009	R041.DTA (smallholder)	ASC 2008/09 (NBS)
	2019	R041_LAND_OWNERSHIP.dta	ASC 2019/20 (NBS)

Appendix A. Per-country variable provenance

For every final variable, the exact source file, source variable(s), value-label codes and construction rule, by country and wave (verified against the raw data).

ETHIOPIA — Ethiopia Socioeconomic Survey (ESS)

1. Survey & wave key

Wave	Round	Year (assigned)	Parcel roster	Field roster	HH cover	HH id
1	ESS 11	2012	sect2_pp_w1.dta	sect3_pp_w1.dta	sect_cover_hh_w1.dta	household_id
2	ESS 13	2014	sect2_pp_w2.dta	sect3_pp_w2.dta	sect_cover_hh_w2.dta	household_id2
3	ESS 15	2016	sect2_pp_w3.dta	sect3_pp_w3.dta	sect_cover_hh_w3.dta	household_id2
4	ESS 18	2019	sect2_pp_w4.dta	sect3_pp_w4.dta	sect_cover_hh_w4.dta	household_id
5	ESS 21	2022	sect2_pp_w5.dta	sect3_pp_w5.dta	sect_cover_hh_w5.dta	household_id

Land-unit conversion file (for self-reported area): ET_local_area_unit_conversion.dta, present in ESS 11/13/15/18; **wave 5 reuses the ESS 18 copy** (none ships with ESS 21).

2. Tenure indicators

All four are built on the **parcel roster** (sect2_pp_w*.dta); key = holder_id parcel_id. "Acquisition" = the parcel-acquisition question; its value labels are in §5.

Variable	Wave	Source file	Source var	Construction
parcel_rentedin	1	sect2_pp_w1.dta	pp_s2q03	inlist(pp_s2q03,3) — Rent
parcel_rentedin	2	sect2_pp_w2.dta	pp_s2q03	inlist(pp_s2q03,3) — Rent
parcel_rentedin	3	sect2_pp_w3.dta	pp_s2q03	inlist(pp_s2q03,3,6) — Rent + Sharecrop-in
parcel_rentedin	4	sect2_pp_w4.dta	s2q05	inlist(s2q05,3,6) — Rent + Sharecrop-in

Variable	Wave	Source file	Source var	Construction
parcel_rentedin	5	sect2_pp_w5.dta	s2q05	inlist(s2q05,3,6) — Rent + Sharecrop-in
parcel_rentedout	1	sect2_pp_w1.dta	pp_s2q10	pp_s2q10==1 — any fields rented out (Yes/No)
parcel_rentedout	2	sect2_pp_w2.dta	pp_s2q10	pp_s2q10==1
parcel_rentedout	3	sect2_pp_w3.dta	pp_s2q10	pp_s2q10==1
parcel_rentedout	4	sect2_pp_w4.dta	s2q13	inlist(s2q13,1,2) — all rented out / all sharecropped out
parcel_rentedout	5	sect2_pp_w5.dta	s2q13	inlist(s2q13,1,2) — all rented out / all sharecropped out
parcel_certificate	1	sect2_pp_w1.dta	pp_s2q04 (+pp_s2q03)	pp_s2q04: 1->1, 2->0; then 0 if pp_s2q03 in {3,4,6}
parcel_certificate	2	sect2_pp_w2.dta	pp_s2q04 (+pp_s2q03)	pp_s2q04: 1->1, 2->0; then 0 if pp_s2q03 in {3,4,6}
parcel_certificate	3	sect2_pp_w3.dta	pp_s2q04 (+pp_s2q03)	pp_s2q04: 1->1, 2->0; then 0 if pp_s2q03 in {3,4,6}
parcel_certificate	4	sect2_pp_w4.dta	s2q03	s2q03: 1->1, 2->0 (document)
parcel_certificate	5	sect2_pp_w5.dta	s2q03	s2q03: 1->1, 2->0 (document)
parcel_purchased	1	sect2_pp_w1.dta	pp_s2q03	**missing** — no purchase category asked
parcel_purchased	2	sect2_pp_w2.dta	pp_s2q03	**missing** — no purchase category asked
parcel_purchased	3	sect2_pp_w3.dta	pp_s2q03	pp_s2q03==7 — Purchased
parcel_purchased	4	sect2_pp_w4.dta	s2q05	s2q05==7 — Purchased
parcel_purchased	5	sect2_pp_w5.dta	s2q05	s2q05==7 — Purchased

3. Parcel area (parcel_area_ha)

Built on the **field roster** (sect3_pp_w*.dta) at field level, then summed to the parcel (collapse (sum) … by(holder_id parcel_id)). Field area = **GPS where measured, else self-reported** (both in ha); no model-based imputation, so the measure is deterministic and reproducible across languages. n_fields = count of fields with a non-missing area. Parcels with no cultivated field (e.g. fully rented out) have missing area and n_fields==0.

Component	Wave	Source file	Source var(s)	Notes
GPS area (primary)	1	sect3_pp_w1.dta	pp_s3q05_a	x 0.0001 -> ha; keep if > 0
GPS area (primary)	2	sect3_pp_w2.dta	pp_s3q05_a	x 0.0001 -> ha
GPS area (primary)	3	sect3_pp_w3.dta	pp_s3q05_a	x 0.0001 -> ha
GPS area (primary)	4	sect3_pp_w4.dta	s3q08 (flag s3q07)	x 0.0001 if s3q07==1
GPS area (primary)	5	sect3_pp_w5.dta	s3q08 (flag s3q07)	x 0.0001 if s3q07 in {1,2}
Self-reported area (fallback when GPS missing)	1–3	sect3_pp_w*.dta	pp_s3q02_a, unit pp_s3q02_c	converted to ha via ET_local_area_unit_conversion.dta
Self-reported area (fallback when GPS missing)	4–5	sect3_pp_w*.dta	s3q02a, unit s3q02b	converted to ha (w5 uses ESS 18 conversion file)

4. Design variables & identifiers

Variable	Wave	Source file	Source var(s)	Construction
`weight`	1	`sect_cover_hh_w1.dta`	`pw`	key `household_id`
`weight`	2	`sect_cover_hh_w2.dta`	`pw2`	key `household_id2`
`weight`	3	`sect_cover_hh_w3.dta`	`pw_w3`	key `household_id2`
`weight`	4	`sect_cover_hh_w4.dta`	`pw_w4`	key `household_id`
`weight`	5	`sect_cover_hh_w5.dta`	`pw_w5`	key `household_id`
`ea_id` (PSU)	1–5	`sect2_pp_w*.dta`	`ea_id`	carried raw from parcel roster
`strataid`	1	`sect_cover_hh_w1.dta`	`rural`, `saq01`	`group(rural region2)`; `region2`: `saq01 in {2,6,12,13,15}`->99
`strataid`	2	`sect_cover_hh_w2.dta` (+ w1 strata)	`rural`, `saq01`	chained from w1; new urban (`rural==3`) strata added (+10)
`strataid`	3	`sect_cover_hh_w3.dta` (+ w2 strata)	`rural`, `saq01`	chained from w2; new HHs = max strata within `saq01xrural`
`strataid`	4	`sect_cover_hh_w4.dta`	`saq14`, `saq01`	explicit regionxrural/urban recode (codes 1–32, 99)
`strataid`	5	`sect_cover_hh_w5.dta`	`saq14`, `saq01`	explicit regionxrural/urban recode (codes 1–32, 99)
`holder_id`, `parcel_id`	1–5	`sect2_pp_w*.dta`	`holder_id`, `parcel_id`	parcel key (also present on field roster)
`hh_id`	1–5	parcel roster / cover	`household_id` (w1,4,5) / `household_id2` (w2,3)	renamed `hh_id`
`year`	1–5	—	—	assigned: 2012/2014/2016/2019/2022
`country`	1–5	—	—	assigned `"Ethiopia"`

5. Value labels of key source variables (verified against the raw rosters)

Parcel acquisition — `pp_s2q03` (w1–3) / `s2q05` (w4–5), *"How did your household acquire [PARCEL]?"*

Code	w1 (ESS11)	w2 (ESS13)	w3 (ESS15)	w4 (ESS18)	w5 (ESS21)
1	Granted by local leaders	Granted by local leaders	Granted by local leaders	Granted by local leaders	Granted by local leaders
2	Inherited	Inherited	Inherited	Gift / Inherited	Inherited
3	**Rent**	**Rent**	**Rent**	**Rent**	**Rent**
4	Borrowed for free	Borrowed for free	Borrowed for free	Borrowed for free	Borrowed for free
5	Moved in w/o permission	Moved in w/o permission	Moved in w/o permission	Moved in w/o permission	Moved in w/o permission
6	Other (specify)	Other (specify)	**Shared crop in**	**Shared crop in**	**Shared crop in**
7	—	—	**Purchased**	**Purchased**	**Purchased**
8	—	—	Other (specify)	Other (specify)	Other (specify)
10/11/12	Moved-in / Other variants	—	—	—	—

Rented out — `pp_s2q10` (w1–3): `1 = Yes` (any fields rented out), `2 = No`. `s2q13` (w4–5): `1 = all rented out`, `2 = all sharecropped out`, `3 = all given out free`, `4 = No`.

Certificate / document — `pp_s2q04` (w1–3, "certificate") / `s2q03` (w4–5, "document"): `1 = Yes`, `2 = No`.

Rural/urban for strata — `rural` (w1–3) and `saq14` (w4–5, `1 = rural`, `2 = urban`, small-town categories per round).

6. Harmonization decisions & caveats (Ethiopia)

- **Unit = parcel.** The universe is every parcel in the parcel roster, so parcels entirely rented/sharecropped out are retained even though they have no cultivated fields. (Basing on the field roster zeroed out rented-out in w4–5 — see project notes.)
- **`parcel_purchased` is missing in w1–w2.** ESS11/ESS13 offered no "Purchased" acquisition category, so purchase is unmeasured (`.`), not a structural zero. *Estimate this variable separately:* a joint `mean/svy: mean` with the other variables uses casewise deletion and silently drops all of 2012–2014.
- **Rental scope: both cash/fixed rental AND sharecropping are counted - read with care across waves.** Rented-in is built from the parcel-acquisition question, where "Rent" (code 3) and "Shared crop in" (code 6) are separate categories; we include both. But the "shared crop in" category only existed from wave 3 (2016) onward - in waves 1-2 (2012, 2014) the questionnaire had no separate sharecropping-in option, so those years capture rent only (any sharecropping-in fell under "Other"). This is part of why rented-in ticks up from 2014 to 2016. Rented-out also includes both, but the explicit split appears only from wave 4 (2019): `s2q13` distinguishes "all rented out" (1) vs "all sharecropped out" (2), and we count both. In waves 1-3 (2012-2016) rented-out comes from a single yes/no item ("were any fields rented out?") that does not separately label sharecropping, so it is captured broadly rather than itemized. Excluded on both sides: "borrowed for free" (acquisition code 4) and "given out for free" (disposal code 3 in 2019/2022) - i.e., free / non-market transfers are not counted as renting. And because "Rent" (code 3) is not itself split into cash vs fixed in-kind rent, "rented in/out" here means cash-or-fixed rental plus sharecropping, not cash-only.
- **Area** is cultivated field area summed to the parcel, using GPS where measured and self-reported area as a fallback where GPS is missing. The published pipeline instead model-imputes missing GPS (pmm); we use the deterministic GPS-else-self-reported measure so area reproduces exactly across Stata/R/Python. Tenure variables are unaffected.
- **ESS21 area** omits the `sect12c` ag-extension self-reported supplement used in the full pipeline, because its source merge key is inconsistent (`household_id+field_id` vs `holder_id+parcel_id+field_id`). w5 area uses the field-roster GPS-else-self-reported measure.
- **Weights** are taken from the household cover file; design-based estimates use `svyset ea_id [pw=weight], strata(strataid)` (PSU/strata interacted with wave when pooling).

MALAWI — Integrated Household Panel Survey (IHPS)

Source: the four-wave IHPS **panel** release `MWI_2010-2019_IHPS_v06`, extracted flat to `Malawi/IHPS_panel_v6/MWI_2010-20` (all waves point at this one folder; filenames carry the year suffix).

> **Unit note.** The survey's land unit changed: in 2010/2013 tenure is asked at the **plot** level (no garden grouping), so `parcel` := plot; in 2016/2019 tenure is asked at the **garden** level and `parcel` := garden (area summed from its plots).

1. Survey & wave key

Wave	Round	Year	HH id	Tenure module (unit)	Area module
1	IHS3/IHPS	2010	`case_id`	`ag_mod_d_10` (plot)	`ag_mod_c_10`
2	IHPS	2013	`y2_hhid`	`ag_mod_d_13` (plot)	`ag_mod_c_13` + `ag_mod_o2_13`
3	IHPS	2016	`y3_hhid`	`ag_mod_b2_16` (garden)	`ag_mod_c_16` + `ag_mod_o2_16`

Wave	Round	Year	HH id	Tenure module (unit)	Area module
4	IHPS	2019/20	y4_hhid	ag_mod_b2_19 (garden)	ag_mod_c_19 + ag_mod_o2_19

Household cover (weight, ea_id, strata): hh_mod_a_filt_<yy>.dta.

2. Tenure indicators

Variable	Wave	Source file	Source var	Construction
parcel_rentedin	1	ag_mod_d_10	ag_d03	inlist(ag_d03,6,7,8) — leasehold/rent/tenant
parcel_rentedin	2	ag_mod_d_13	ag_d03	inlist(ag_d03,6,7,8)
parcel_rentedin	3	ag_mod_b2_16	ag_b203	inlist(ag_b203,6,7,8)
parcel_rentedin	4	ag_mod_b2_19	ag_brentedin, ag_b211a/b	ag_brentedin==1 OR paid owner ag_b211a/b>0 (no acq. question in 2019)
parcel_rentedout	1-2	ag_mod_d_<yy>	ag_d19a-d	rent received >0 (cash/in-kind, already/still)
parcel_rentedout	3	ag_mod_b2_16	ag_b219a-d	rent received >0
parcel_rentedout	4	ag_mod_b2_19	ag_brentedout, ag_b219a-d	ag_brentedout==1 OR ag_b219a-d>0
parcel_certificate	1	—	—	**missing** — not asked
parcel_certificate	2	ag_mod_d_13	ag_d03_1	ag_d03_1==1 (has title)
parcel_certificate	3	ag_mod_b2_16	ag_b204_1	inlist(ag_b204_1,1,2,3) (lease offer / title deed / lease cert)
parcel_certificate	4	—	—	**missing** — not asked
parcel_purchased	1-2	ag_mod_d_<yy>	ag_d03	inlist(ag_d03,4,5) — purchased w/ or w/o title
parcel_purchased	3	ag_mod_b2_16	ag_b203	ag_b203==4 — purchased
parcel_purchased	4	—	—	**missing** — acquisition question dropped in 2019

3. Parcel area (parcel_area_ha)

Field roster ag_mod_c_<yy> (+ perennial ag_mod_o2_<yy> for w2-4). Field area = GPS (ag_c04c) where measured, else self-reported (ag_c04a, unit ag_c04b: 1=acre, 2=ha, 3=m²); acres->ha via x0.404686. No model-based imputation (the published pipeline pmm-imputes missing GPS; we use the deterministic GPS-else-self-reported measure for cross-language reproducibility). Summed to the parcel: garden in w3-4; in w1-2 each plot is its own parcel. n_fields = cultivated fields per parcel.

4. Design variables & identifiers

Variable	Wave	Source file	Source var(s)	Construction
weight	1	hh_mod_a_filt_10	hh_wgt	baseline sampling weight
weight	2	hh_mod_a_filt_13	panelweight	panel weight 2013
weight	3	hh_mod_a_filt_16	panelweight_2016	panel weight 2016
weight	4	hh_mod_a_filt_19	panelweight_2019	panel weight 2019
ea_id	1-4	hh_mod_a_filt_<yy>	ea_id	enumeration area (PSU)
strataid	1-2	hh_mod_a_filt_<yy>	stratum	baseline stratum (region x urban/rural)

Variable	Wave	Source file	Source var(s)	Construction
`strataid`	3-4	`hh_mod_a_filt_<yy>`	`region`, `reside`	`group(region reside)` (no `stratum` in cover)
`parcel_id`	1-2	tenure module	`hh_id` + plot no. (`ag_d00`)	concatenated
`parcel_id`	3-4	tenure module	`hh_id` + `gardenid`	concatenated
`year`	1-4	—	—	2010 / 2013 / 2016 / 2019

5. Value labels of key source variables (verified)

Acquisition — `ag_d03` (w1-2) / `ag_b203` (w3): *"How did your household acquire this [plot/garden]?"*

Code	w1-2 (ag_d03)	w3 (ag_b203)
1	Granted by local leaders	Granted by local leaders
2	Inherited	Inherited
3	Bride price	Bride price
4	**Purchased (with title)**	**Purchased**
5	**Purchased (no title)**	—
6	Leasehold	Leasehold
7	Rent short-term	Rent short-term
8	Farming as a tenant	Farming as a tenant
9	Borrowed for free	Borrowed for free
10	Moved in w/o permission	Moved in w/o permission
11	Other	Other
12 / 13	—	Allocated by family / Gift from non-HH

(Wave 4 has no acquisition-method question — only "from whom" and "year acquired".)

Rented out / in (w4 flags) — `ag_brentedin` ("gave output as rent" -> rented in), `ag_brentedout` ("received output as rent" -> rented out): `1 = Yes`, `2 = No`. Rent amounts `ag_d19a-d` / `ag_b219a-d` = cash/in-kind received (already / still due).

Certificate — `ag_d03_1` (w2, has title): `1 = Yes`, `2 = No`. `ag_b204_1` (w3): `1` offer of lease · `2` title deed · `3` certificate of lease · `4` no · `96` other.

6. Harmonization decisions & caveats (Malawi)

- **Rental variables were derived here, not inherited** — the upstream pipeline never built `plot_rentedin/out` for Malawi. They are constructed from the acquisition question (rented-in) and the rent-received variables (rented-out).
- **Land unit changes across waves** — plot (2010/2013) vs garden (2016/2019). Counts and mean areas are not strictly unit-comparable across that break.
- **Wave 4 (2019)**: the categorical acquisition question was dropped, so `parcel_purchased` and `parcel_certificate` are **missing** in 2019, and rented-in uses a payment-based proxy (`ag_brentedin` / paid-owner) rather than an acquisition code.
- **Rented-in** = leasehold/rent/tenant (codes 6,7,8); "borrowed for free" (9) is excluded as non-market access.
- **Rented-out is NOT comparable across waves - do not read the trend.** It is captured differently by wave: 2010 and 2013 identify rented-out only via positive rent *received* (`ag_d19a-d`), with heavy skip/missing on those amount items, whereas 2019 has an explicit yes/no flag (`ag_brentedout`). As a result the weighted rate is ~0.2% in 2010 versus ~1.3-1.5% in 2016/2019. The early levels almost

certainly understate rented-out, so the apparent increase over time is largely a measurement artifact rather than a real change in behavior.

- **Purchase** is genuinely measurable here (codes 4-5), unlike Ethiopia w1-2.
- **Strata**: baseline `stratum` (w1-2) vs `group(region reside)` (w3-4).
- **Year map** follows the IHPS rounds (2010/2013/2016/2019); the shocks do-file used 2017/2020 for w3/w4 — change `year` in `extract_MWI.do` if you prefer that labeling.

MALI — Enquete Agricole de Conjoncture Integree (EACI)

Two separate cross-sectional rounds (not a panel): EACI 2014 and EACI 2017. The 2017 files use a different naming scheme (`eaci17_sNNpY`) and variable prefix (`s11b…`) than 2014 (`EACI…_p1`, `s1b…`).

> **Unit note.** Everything we use - tenure, acquisition, disposition, and area - sits in the single parcel/exploitation roster, so the `parcel` is that roster record and no cross-module merge is needed.

1. Survey & wave key

Wave	Round folder	Year	HH id	Parcel roster
1	`Mali/EACI 14`	2014	`grappe-menage`	`EACIEXPLOI_p1.dta` (`s1b…`)
2	`Mali/EACI 17`	2017	`grappe-exploitation`	`eaci17_s11bp1.dta` (`s11b…`)

Weights: 2014 `EACIPOIDS.dta` (`poids_menage`); 2017 `EACI17_ECHANTILLON.dta` (`poids_leger`, and the official `strate`). Cover (2014 strata): `EACICONTROLE_p1.dta`.

2. Tenure indicators

Variable	Wave	Source var	Construction
`parcel_rentedin`	1	`s1bq17`	`inlist(s1bq17,4,5)` - Location + Metayage
`parcel_rentedin`	2	`s11bq17`	`inlist(s11bq17,4,5)` - Location + Metayage
`parcel_certificate`	1	`s1bq17`	`s1bq17==1` - owned with formal title
`parcel_certificate`	2	`s11bq17`	`s11bq17==1` - owned with formal title
`parcel_purchased`	1	`s1bq22`	`s1bq22==7` - Achat
`parcel_purchased`	2	`s11bq22`	`s11bq22==7` - Achat
`parcel_rentedout`	1-2	—	**missing** - not measurable (see §6)

3. Parcel area (`parcel_area_ha`)

GPS where measured, else self-reported (deterministic; no imputation). 2014: GPS `s1bq05a`, self-reported `s1bq10` (both treated as ha; value 99 = missing). 2017: GPS `s11bq07`, self-reported `s11bq11a` (x 0.0001 where unit `s11bq11b==2`). `n_fields` = parcel records aggregated to the parcel id (typically 1).

4. Design variables & identifiers

Variable	Wave	Source	Construction
`weight`	1	`EACIPOIDS.dta`	`poids_menage`
`weight`	2	`EACI17_ECHANTILLON.dta`	`poids_leger`
`ea_id`	1-2	parcel roster	`grappe` (cluster / PSU)
`strataid`	1	`EACICONTROLE_p1.dta`	`group(s00q01 s00q04)` (region x milieu)

Variable	Wave	Source	Construction
`strataid`	2	`EACI17_ECHANTILLON.dta`	official `strate`
`parcel_id`	1	parcel roster	`grappe-menage-s1bq01-s1bq02`
`parcel_id`	2	parcel roster	`grappe-exploitation-s11bq01-s11bq02`
`year`	1-2	—	2014 / 2017

5. Value labels of key source variables (verified)

Occupation mode - `s1bq17` (2014) / `s11bq17` (2017), *"Mode d'occupation / propriete de la parcelle"*: 1 Propriete avec titre · 2 Propriete sans titre · 3 Pret gratuit · **4 Location** · **5 Metayage** · 6 Gage · 7 Autre · (9 / 99 = missing).

Acquisition mode - `s1bq22` (2014) / `s11bq22` (2017), *"Mode d'acquisition de la parcelle"*: 1 Heritage · 2 Par mariage · 3 Attribution coutumiere · 4 Don · 5 Attribution ODR · 6 Appropriation · **7 Achat** · 8 Autre.

Disposition - `s1bq32` (2014): 1 En jachere, 2 Exploitee, 9 missing (no rented-out code). `s11bq32` (2017): 1 Jachere, 2 Louee/Pretee, 3 Exploitee.

6. Harmonization decisions & caveats (Mali)

- **Rental variables derived here, not inherited** - the upstream pipeline built only `plot_owned`/`plot_certificate` for Mali. Rented-in and purchased are constructed from the occupation (`*q17`) and acquisition (`*q22`) questions.
- **Rented-out is NOT measurable in EACI (missing both waves).** The roster covers the parcels a household *operates* (its exploitation), so land rented/lent OUT is out of frame. 2014 has no rented-out code in the disposition item; 2017's "Louee/Pretee" code flags only 5 of ~24,250 parcels - a structural undercount, not a real ~0% rate.
- **Rented-in** = Location (4) + Metayage (5); "Pret gratuit" (3, borrowed free) and "Gage" (6, pledge/collateral) are excluded as non-market arrangements.
- **Certificate** here means *owned with a formal title* (`*q17==1`), the closest EACI analog; a non-owned parcel cannot carry a household title.
- **Purchase is measurable in both waves** (acquisition code 7 = Achat) - unlike Ethiopia (w1-2) and Malawi (2019).
- **Strata**: 2017 uses the official `strate`; 2014 has none in its own files, so we build `group(region milieu)` = `group(s00q01 s00q04)` from the cover as a self-contained design-stratum proxy.
- **Cross-wave comparability - read levels, not the 2014->2017 change.** EACI 2014 and 2017 are *independent cross-sections* (not a panel) with very different sample sizes and frames: 2,299 households / 9,658 parcels in 2014 vs 6,293 / 24,250 in 2017, the latter even more dominated by owner-cultivators (untitled ownership 87% -> 93% of parcels). The *absolute* count of rented-in parcels is essentially unchanged (116 -> 106), so the fall in the rented-in *rate* (1.3% -> 0.4%) is mechanical dilution by a larger, more owner-heavy denominator, not a behavioral decline. Rental is also a rare event (~100 parcels/wave), so the estimates carry wide confidence intervals - check CI overlap before interpreting any cross-wave difference. The robust takeaway is that Mali's land-rental market is thin (<2% of parcels), consistent with customary tenure.
- **Encoding**: the 2017 files are Latin-1 (French accents); numeric codes are unaffected, but read with the right encoding outside Stata (e.g. `encoding="latin1"`).

NIGER — Enquete Nationale sur les Conditions de Vie (ECVMA)

Two ECVMA rounds (2011, 2014). The 2014 round uses **uppercase** variable names (`AS01Q*`) and a 3-part household id; 2011 uses lowercase (`as01q*`) and a single `hid`. Some files are Latin-1 encoded.

Unit note. Tenure, acquisition, title, disposition, and area all sit in the one parcel roster, so the parcel is that roster record and no cross-module merge is needed.

1. Survey & wave key

Wave	Round folder	Year	HH id	Parcel roster
1	Niger/ECVMA 11	2011	hid	ecvmaas1_p1.dta (as01q*)
2	Niger/ECVMA 14	2014	GRAPPE-MENAGE-EXTENSION	ECVMA2_AS1P1.dta (AS01Q*)

Weights: 2011 ecvmamen_p1_en.dta (housing; hhweight, grappe, strate); 2014 ECVMA2014_P1P2_ConsoMen.dta (hhweight) + cover ECVMA2_MS00P1.dta (region).

2. Tenure indicators

Variable	Wave	Source var	Construction
parcel_rentedin	1	as01q16	as01q16==4 - Location (rental)
parcel_rentedin	2	AS01Q14	AS01Q14==3 - Location (rental)
parcel_purchased	1	as01q19	as01q19==1 - Achat (9 -> missing)
parcel_purchased	2	AS01Q18	AS01Q18==1 - Achat (9 -> missing)
parcel_certificate	1	as01q18	inlist(as01q18,1,2,3,4) - any title doc (9 -> missing)
parcel_certificate	2	AS01Q15	inlist(AS01Q15,1,2,3,4) - any title doc (9 -> missing)
parcel_rentedout	1	as01q41	as01q41==4 - "louee a un autre" (rented out)
parcel_rentedout	2	AS01Q39	AS01Q39==4 - "louee a un autre" (rented out)

3. Parcel area (parcel_area_ha)

GPS where measured, else self-reported (deterministic; no imputation). 2011: GPS as01q09, self-reported as01q08; 2014: GPS AS01Q07, self-reported AS01Q06. All in m^2 (x 0.0001 -> ha); 999999 = missing; GPS 0 = missing.

4. Design variables & identifiers

Variable	Wave	Source	Construction
weight	1	ecvmamen_p1_en.dta	hhweight
weight	2	ECVMA2014_P1P2_ConsoMen.dta	hhweight
ea_id	1	ecvmamen_p1_en.dta	grappe (cluster / PSU)
ea_id	2	ECVMA2_MS00P1.dta	GRAPPE
strataid	1	ecvmamen_p1_en.dta	official strate
strataid	2	ECVMA2_MS00P1.dta	MS00Q01 (region; proxy - no 2014 strate)
parcel_id	1	parcel roster	hid-as01q03-as01q05
parcel_id	2	parcel roster	GRAPPE-MENAGE-EXTENSION-AS01Q01-AS01Q03
year	1-2	—	2011 / 2014

5. Value labels of key source variables (verified)

Occupation mode - as01q16 (2011) / AS01Q14 (2014), *"Mode d'occupation de la parcelle"*: 2011: 1 propriete · 2 pret (gratuit) · 3 hypotheque (gage) · **4 location** · 5 autres. 2014: 1 propriete · 2 copropriete · **3 location** · 4 hypotheque · 5 pret · 6 autres. *(No sharecropping / metayage category in either wave.)*

Acquisition mode - `as01q19` / `AS01Q18`: **1 Achat** · 2 Heritage · 3 Don · 4 Prise de possession simple · 5 Autres · 9 manquant.

Title document - `as01q18` / `AS01Q15`: 1 titre foncier · 2 certificat coutumier · 3 attestation de vente · 4 autre document · 5 aucun document · 9 manquant.

Disposition (reason for non-cultivation) - `as01q41` / `AS01Q39`: 1 jachere · 2 prete a un autre · 3 hypothequee · **4 louee a un autre** · 5 autre · 9 manquant.

6. Harmonization decisions & caveats (Niger)

- **Rental variables derived here, not inherited** - upstream built only `plot_owned/plot_certificate`. Rented-in (Location), purchased (Achat) and rented-out (louee a un autre) are constructed from the occupation, acquisition and disposition questions.
- **Rented-in = rental (Location) only** - Niger has no sharecropping category, so unlike Ethiopia/Malawi/Mali the rented-in measure excludes sharecropping. Free loan (pret) and pledge (hypotheque) are also excluded as non-market arrangements.
- **Rented-out UNDERCOUNTS - treat as a lower bound.** It is captured only through the "reason for non-cultivation" item (`as01q41==4`), so parcels rented out are largely under-represented in this operated-parcel roster (weighted rate ~0.0% in 2011, ~0.2% in 2014). The code is real (unlike Mali), but the level is not reliable.
- **Cross-wave comparability - do not read the rented-in change.** 2011 and 2014 are independent cross-sections with different occupation-mode category schemes and samples; the rented-in rate falls from ~14% (2011) to ~2.4% (2014), a swing too large to be behavioral and not directly comparable across the redesign. Check CI overlap.
- **Certificate** = the parcel has any tenure document (`title in {1,2,3,4}`); "aucun document" (5) = 0, "manquant" (9) = missing.
- **Strata**: 2011 uses the official `strate`; 2014 has no `strate` in its files, so we use region (`MS00Q01`) as a coarser self-contained stratum proxy.

NIGERIA — General Household Survey-Panel (GHS)

Five GHS-Panel waves (2010/11, 2012/13, 2015/16, 2018/19, 2023). Plot id = `hhid-plotid`. Tenure (acquisition, rented-out, certificate) is in the tenure module; area is in the plot roster - the two are merged on `hhid-plotid`.

> **Unit note.** The `parcel` is the plot record. Tenure variable names shift: wave 1 uses `s11bq*`; waves 2-5 use `s11b1q*`.

1. Survey & wave key

Wave	Round folder	Year	Tenure module	Plot roster
1	`Nigeria/GHS 10`	2011	`sect11b_plantingw1` (`s11bq*`)	`sect11a1_plantingw1`
2	`Nigeria/GHS 12`	2013	`sect11b1_plantingw2` (`s11b1q*`)	`sect11a1_plantingw2`
3	`Nigeria/GHS 15`	2016	`sect11b1_plantingw3`	`sect11a1_plantingw3`
4	`Nigeria/GHS 18`	2019	`sect11b1_plantingw4`	`sect11a1_plantingw4`
5	`Nigeria/GHS 23`	2023	`sect11b1_plantingw5`	`sect11a1_plantingw5`

2. Tenure indicators

Variable	Wave	Source var	Construction
`parcel_purchased`	1	`s11bq4`	`s11bq4==1` - outright purchase

Variable	Wave	Source var	Construction
`parcel_purchased`	2-5	`s11b1q4`	`s11b1q4==1` - outright purchase
`parcel_rentedin`	1	`s11bq4`	`s11bq4==2` - rented for cash/in-kind
`parcel_rentedin`	2-4	`s11b1q4`	`s11b1q4==2`
`parcel_rentedin`	5	`s11b1q4`	`inlist(s11b1q4,2,6)`
`parcel_rentedout`	1	`s11bq17`	`s11bq17==2` - main use = rented out
`parcel_rentedout`	2-4	`s11b1q28`	`inlist(s11b1q28,2,3)`
`parcel_rentedout`	5	`s11b1q44`	`inlist(s11b1q44,2,3)`
`parcel_certificate`	1	—	**missing** - not asked
`parcel_certificate`	2-4	`s11b1q7`	`s11b1q7==1` (0 if not owned)
`parcel_certificate`	5	`s11b1q8`	`s11b1q8==1` (0 if not owned)

3. Parcel area (`parcel_area_ha`)

GPS where measured (the "GPS measured in square meters" variable x 0.0001 -> ha), else self-reported. The GPS variable name shifts by wave: **w1 `s11aq4d`, w2-4 `s11aq4c`, w5 `s11mq3`** (there is no `plot_area` variable in the raw - the upstream pipeline referenced a nonexistent one, so its GPS was effectively unused). Self-reported is in local units with **region-specific** conversion factors (heaps/ridges/stands x the survey zone `admin_1` 1-6; plus acres/m²/plot factors); self-reported number = `s11aq4a` (w1-3) / `s11aq4aa` (w4) / `s11aq3_number` (w5), unit `s11aq4b` (w1-4) / `s11aq3_unit` (w5). The region factors are identical across waves (transcribed from the upstream pipeline).

4. Design variables & identifiers

Variable	Wave	Source	Construction
`weight`	1-3	`cons_agg_waveN_visit1/2.dta`	`hhweight`
`weight`	4	`totcons_final.dta`	`wt_wave4`
`weight`	5	`secta_plantingw5.dta`	`wt_cross_wave5`
`ea_id`	1-5	`secta_planting*` (+ GHS10 fallback)	`lga-ea`
`strataid`	1-5	`secta_planting*` (+ GHS10 fallback)	`zone`
`parcel_id`	1-5	tenure / roster	`hhid-plotid`
`year`	1-5	—	2011 / 2013 / 2016 / 2019 / 2023

5. Value labels of key source variables (verified for GHS10; consistent instrument)

Acquisition - `s11bq4` (w1) / `s11b1q4` (w2-5), *"How was [PLOT] acquired?"*: **1 outright purchase** · **2 rented for cash/in-kind** · 3 used free of charge · 4 distributed by community/family · (w3+ add code 5 = other ownership; w5 adds further non-owned codes incl. 6 = another rental arrangement).

Rented-out - w1 `s11bq17` ("main use of plot"): 1 left fallow, 2 rented out. w2-4 `s11b1q28` / w5 `s11b1q44` (codes 2,3 = rented/leased out).

Certificate - `s11b1q7` (w2-4) / `s11b1q8` (w5): 1 yes, 2 no, 3 = missing.

6. Harmonization decisions & caveats (Nigeria)

- **Purchase recovered from the acquisition question** (`*q4==1`, outright purchase) - the upstream pipeline built owned/rented but not purchase. Verified against the GHS10 value labels; applied to later waves under the consistent GHS instrument (the owned recode `{1,4}`/`{1,4,5}` is consistent with 1 = purchase across waves).
- **Rented-out likely UNDERCOUNTS in wave 1** - it comes from the "main use of plot" item, and since most plots are cultivated, only ~0.15% are flagged rented out. Waves 2-5 use a dedicated rented-out item (`s11b1q28`/`s11b1q44`) that should capture more; treat the wave-1 level as a lower bound and mind cross-wave comparability.

- **Certificate not asked in wave 1** -> missing in 2011 (added from wave 2); coded 0 for non-owned parcels in later waves (a certificate applies to owned land).
- **Area** uses GPS (`plot_area` x 0.0001) else self-reported with Nigeria's region-specific local-unit conversions; deterministic (no imputation).
- **Strata/PSU**: `strataid = zone`, `ea_id = lga-ea`, taken from each wave's cover with the GHS10 cover as a panel fallback (the panel covers drop ea/lga for non-refresh HHs).
- **Most complex country** - 5 waves, region-specific area conversions, panel geography. The first Stata run is a shakedown: confirm the area GPS source (`plot_area`) and the per-wave cover/weight merges resolve as expected.

TANZANIA — National Panel Survey (NPS)

Five NPS waves. Waves **4 (2014/15) and 5 (2019/20) each split into an EXTENDED (long panel) and a REFRESH subsample** - 7 datasets in all, combined here into waves 4 and 5 using each subsample's own weights. Tenure (acquisition & use) is in the plot-inputs module; area is in the plot roster - merged on `hhid-plotnum`.

> **Unit note.** `parcel` = the plot. The household id changes each wave (`hhid` -> `y2_hhid` -> `y3_hhid` -> `y4_hhid` -> `sdd_hhid`/`y5_hhid`).

1. Survey & wave key

Wave	Round folder(s)	Year	HH id	Tenure var
1	`Tanzania/NPS 08`	2009	`hhid`	`s3aq22` (early)
2	`Tanzania/NPS 10`	2011	`y2_hhid`	`ag3a_24` (early)
3	`Tanzania/NPS 12`	2013	`y3_hhid`	`ag3a_25` (early)
4	`NPS 14 - extended` + `NPS 14 - refresh`	2015	`y4_hhid`	`ag3a_25` (late)
5	`NPS 19 - extended` (`sdd_hhid`) + `NPS 19 - refresh` (`y5_hhid`)	2019	—	`ag3a_25` (late)

Plot roster: `SEC_2A/AG_SEC2A/AG_SEC_2A/AG_SEC_02`. Weights from the cover (`hh_weight/y2_weight/y3_weight/y4_weig`

2. Tenure indicators

Variable	Wave	Source var	Construction
`parcel_rentedin`	1-3	`s3aq22/ag3a_24/ag3a_25`	`inlist(.,3,4)` - rented in + shared-rent
`parcel_rentedin`	4-5	`ag3a_25`	`inlist(ag3a_25,7,8)` - rented in + shared-rent
`parcel_rentedout`	1-5	`s3aq3/ag3a_03`	`== 2` (plot "rented out" in the use question)
`parcel_purchased`	1-3	—	**missing** - no acquisition/purchase category
`parcel_purchased`	4-5	`ag3a_25`	`ag3a_25==5` - PURCHASED
`parcel_certificate`	1	`s3aq25`	`==1` (has title)
`parcel_certificate`	2	`ag3a_27`	`==1`
`parcel_certificate`	3	`ag3a_28`	not in {9,10,11} = has title
`parcel_certificate`	4-5	`ag3a_28a`,`ag3a_28d`	`ag3a_28a in {1,2}` or `ag3a_28d in 1-5`

(Certificate set to 0 for non-owned parcels.)

3. Parcel area (`parcel_area_ha`)

GPS where measured, else self-reported (deterministic; both in **acres x 0.404686 -> ha**). Self-reported = `s2aq4` (w1) / `ag2a_04` (w2-5); GPS = `area` (w1) / `ag2a_09` (w2-5).

4. Design variables & identifiers

Variable	Wave	Source	Construction
`weight`	1-5	cover (`HH_SEC_A/SEC_A_T`)	wave-specific weight; extended & refresh use their own
`strataid`	1-5	cover	`strataid` (direct)
`ea_id`	1-5	cover	cluster/admin vars (region-district-ward-ea), else household id
`parcel_id`	1-5	tenure/roster	`hhid-plotnum`
`year`	1-5	—	2009 / 2011 / 2013 / 2015 / 2019

5. Value labels of key source variables (verified)

EARLY ownership status - `s3aq22/ag3a_24/ag3a_25` (w1-3): 1 owned · 2 used free · **3 rented in** · **4 shared rent** · 5 shared ownership. *(No purchase category.)*

LATE "How was this plot acquired?" - `ag3a_25` (w4-5): 1 inheritance · 2 gift · 3 borrowing · 4 village allocation · **5 PURCHASED** · 6 used free · **7 rented in** · **8 shared-rent** · 9 shared-own · 10 squatting · 11 other.

Use of plot - `s3aq3/ag3a_03`: 1 cultivated · **2 rented out** · 3 given out · 4 fallow · 5 forest · 6 other.

6. Harmonization decisions & caveats (Tanzania)

- **Purchase only measurable from 2015.** NPS1-3 ask "ownership status" (owned/free/rented/shared) with no acquisition mode, so `parcel_purchased` is missing in 2009/2011/2013. From NPS4 the question becomes "how was this plot acquired?" with a PURCHASED code (5). The 2015 level is high (~25% in the extended panel) - a *stock* ("ever acquired by purchase"); scrutinize and mind the level break from the redesign.
- **Rented-in includes sharecropping** (shared-rent: code 4 early / 8 late) alongside cash rental (3 / 7); the upstream pipeline used pure rental only. Shared-OWN (5 / 9) is treated as owned, not rented.
- **Waves 4 and 5 pool an extended panel and a refresh sample**, each weighted by its own design weight; together they represent the wave's population.
- **Rented-out is well-measured** (the roster lists plots owned *or* cultivated, and the "use" question flags rented-out) - unlike Mali/Niger.
- **ea_id** is built from the cover's cluster/admin variables where present, else the household id (a conservative PSU); `strataid` comes directly from the cover.

UGANDA — National Panel Survey (UNPS)

Seven UNPS rounds (2009/10, 2010/11, 2011/12, 2013/14, 2015/16, 2018/19, 2019/20; internal waves 1-5, 7, 8). The land roster is split into **two modules per round**: `AGSEC2A` = parcels the household **owns / holds use-rights to**, and `AGSEC2B` = parcels **accessed for use but not owned** (rented or borrowed in). These are disjoint parcel sets, so the extractor **stacks** them (origin tagged in `parcel_id`: `-A-` owned, `-B-` accessed) rather than merging them on the parcel number.

> **Unit note.** `parcel` = a parcel record from 2A or 2B. Variable NAMES drift across rounds (`a2aq*/A2aq*/s2aq*`) - notably the **certificate** item moves `A2aq25` (2009/10) -> `a2aq23` (2011-

2015) -> `s2aq23` (2018/19). The household key also changes (`Hhid`/`HHID` numeric -> `hh`/`hhid` string -> 32-char interview GUID).

Season note. The parcel roster (tenure, certificate, purchase, area) is **annual** and identical across the two cropping seasons; **only `parcel_rentedout` is season-specific** (read from the per-season "primary use of parcel" item, code 3). The extractor writes one row per parcel **x season** (`season` = 1, 2) differing only in `parcel_rentedout`, and **the paper reports season 1**. *The season order is SWAPPED in 2015, 2018 and 2019:* there the "…b" primary-use variable is the 1st season and "…a" the 2nd (handled per wave).

1. Survey & wave key

Wave	Round folder	Year	HH key (roster <-> cover)	Cross-sec weight
1	`Uganda/UNPS 09`	2009	`Hhid` <-> `HHID`	`wgt09`
2	`Uganda/UNPS 10`	2010	`HHID` <-> `HHID`	`wgt10`
3	`Uganda/UNPS 11`	2011	`HHID` <-> `HHID`	`mult`
4	`Uganda/UNPS 13`	2013	`hh` <-> `hhid` (string)	`wgt_X`
5	`Uganda/UNPS 15`	2015	`hhid` <-> `hhid`	`h_xwgt_W5`
7	`Uganda/UNPS 18`	2018	`hhid` <-> `hhid` (GUID)	`wgt`
8	`Uganda/UNPS 19`	2019	`hhid` <-> `hhid` (GUID)	`wgt`

Land roster: `AGSEC2A` (owned) + `AGSEC2B` (accessed). Cover: `GSEC1`. Household keys and their types were verified against the raw files (roster numeric ids are matched to the string cover ids via `tostring … , format("%17.0f")`, as in the upstream pipeline).

2. Tenure indicators

Variable	Module	Source var (across waves)	Construction
`parcel_rentedin`	2B	rent paid `A2bq9`/`a2bq9`/`a2bq09`	`> 0` (accessed parcel paid for)
`parcel_rentedin`	2A	acquisition `A2aq8`/`a2aq8`/`s2aq8`	`== 3` (Leased-in)
`parcel_rentedout`	2A/2B	per-season primary use	`== 3` ("Rented-out" 2A / "Sub-contracted out" 2B)
`parcel_purchased`	2A	acquisition `A2aq8`/`a2aq8`/`s2aq8`	`== 1` Purchased (`.` in 2018)
`parcel_certificate`	2A	`A2aq25`(09/10) / `a2aq23`(11-15) / `s2aq23`(18/19)	`inlist(1,2,3)`=1; 4=0; accessed parcels=0

Primary-use **season-1** variable by wave: `A2aq13a`(09) · `a2aq13a`(10) · `a2aq11a`(11,13) · `a2aq11b`(15) · `s2aq11b`(18,19) — swapped from 2015. The 2B analogues are `A2bq15*`/`a2bq15*`/`a2bq12*`.

3. Parcel area (`parcel_area_ha`)

GPS where measured, else self-reported (deterministic; both in **acres x 0.404686 -> ha**). GPS = `A2aq4`/`a2aq4`/`s2aq4` (2A), `A2bq4`/`a2bq4`/`s2aq04` (2B); self-reported = the matching `…5`/`…05` variable. Top-coded at `${area_max}` ha.

4. Design variables & identifiers

Variable	Source	Construction
`weight`	cover `GSEC1`	cross-sectional HH weight (see wave key)
`strataid`	cover	`group()` of `stratum` (09/10) / `sregion` (11-15) / `region` (18/19)
`ea_id`	cover	`comm` (09-11) / `ea` (13/15); **blank in 2018/19** (no EA on the cover)
`parcel_id`	roster	`hh_id + "-A-"/"-B-" + parcelID` (origin-tagged, globally unique)
`year`	—	2009 / 2010 / 2011 / 2013 / 2015 / 2018 / 2019

5. Value labels of key source variables (verified against the raw rosters)

Acquisition (2A) `…aq8`: **1 Purchased** · 2 inherited / received as gift · **3 Leased-in** · 4 just walked in (cleared) · 5 don't know · 6 other (2018/19 add 7 given by gov / authority · 8 agreement with owner · 9 without agreement · 96 other). *(2018: entirely missing.)*

Certificate / title (2A) `…aq25/…aq23/s2aq23`: **1 certificate of title** · **2 certificate of customary ownership** · **3 certificate of occupancy** · 4 no document.

Primary use of parcel (per season): 1 own-cultivated annual · 2 own-cultivated perennial · **3 Rented-out** (2A) / **3 Sub-contracted out** (2B) · 4 cultivated by mailo tenant (2A) · 5 fallow · 6 pasture / grazing · 7 woodlot / forest · 96 other.

Tenure system `…aq7`: 1 freehold · 2 leasehold · 3 mailo · 4 customary · 6 other. *(Recorded but not used directly; rented-in keys off rent-paid / leased-in acquisition.)*

6. Harmonization decisions & caveats (Uganda)

- **Owned vs accessed are separate modules.** 2A (owned) supplies purchase, certificate and rented-out; 2B (accessed) supplies rented-in. They are stacked into one parcel frame; accessed parcels carry `parcel_purchased = 0` and `parcel_certificate = 0` (the household holds no title to a rented-in parcel), so the rates use an all-parcels denominator comparable to the other countries.
- **Rented-in = paid access.** A 2B parcel counts as rented-in when rent paid `> 0` (cash or sharecrop); free-borrowed parcels (rent 0) are not rented-in. 2A parcels with acquisition "Leased-in" (code 3) are also flagged rented-in.
- **Rented-out is a lower bound.** It is read from the categorical per-season primary use ("Rented-out" / "Sub-contracted out"); weighted rates are low (~0.2-1.3 %), as in Mali / Niger. The cash "rent received" amount (`…aq14/s2aq14`) corroborates but is not used to define the flag (it is annual, not season-specific).
- **2018 is a reduced panel re-interview.** UNPS7 (2018/19) re-asked owned parcels with a shortened update instrument: most parcel characteristics were carried forward, not re-collected. In `AGSEC2A` the acquisition item `s2aq8` has **0** non-missing values (so `parcel_purchased` is set `.` - do not read the 2B-only zeros as 0% purchase), and GPS / self-reported **area** were captured for only ~256 / ~824 of 4,368 parcels (~25%); the tenure-system item likewise (~932). Certificate (`s2aq23`), ownership flag, documents, disputes and value WERE re-asked. **Consequence:** 2018 owned-parcel area and tenure-system rest on a non-random ~25% (new / re-measured parcels), so treat the 2018 area mean with caution; rented-in (from the separately-administered `AGSEC2B`), rented-out (primary use) and certificate are well-populated. 2019 (UNPS8) is a full re-administration.
- **Season order swapped in 2015 / 2018 / 2019** (the "…b" primary-use variable is season 1); only `parcel_rentedout` differs across seasons, and the paper reports season 1.
- **Deterministic area** (GPS else self-reported, no pmm imputation, unlike the upstream pipeline); acres -> ha via 0.404686; top-coded at 40 ha.
- **Weights** are each round's **cross-sectional** household weight; `ea_id` (PSU) is absent on the 2018/2019 cover, so design-based SEs there rest on strata + weights (point estimates are unaffected). Validated

by a Python replication of the extractor against the raw data: household-weight match 97-100 % and rates within expected ranges for all waves.

ZAMBIA — Rural Agricultural Livelihoods Survey (RALS) *(non-LSMS-ISA)*

Three RALS rounds (2012, 2015, 2019), IAPRI / Indaba Agricultural Policy Research Institute - a **non-LSMS** national agricultural panel. The land roster is the per-field file `field.dta` in every round, and it **includes rented-out and borrowed-out fields**, so rented-out is observed at field level (unlike Mali/Niger). One agricultural season only, so `season` = 1.

> **Unit note.** `parcel` = a **field** (`cluster-hh-field`); single level. Variable names are UPPERCASE in 2012 (`F01`, `F05`, `F06` …) and lowercase from 2015 (`f01` …).

1. Survey & wave key

Wave	Round folder	Year	Field roster	HH/field key	Cross-sec weight
1	`Zambia/RALS 12`	2012	`field.dta` (UPPER)	`cluster-hh-field`	`weight` (from `id.dta`)
2	`Zambia/RALS 15`	2015	`field.dta` (lower)	`cluster-hh-field`	`popwgt` (on `field.dta`)
3	`Zambia/RALS 19`	2019	`field.dta` (lower)	`cluster-hh-field`	`weight` = panel (no popwgt)

Field-type roster (household counts of each field type): `fieldtype.dta` (2012/2019), `allfields.dta` (2015) - not used for the headline rates (the per-field `field.dta` already carries rented-out), but available for cross-checks.

2. Tenure indicators

Variable	Source var	Construction
`parcel_rentedin`	`F01` land use	`== 2` ("rented in", cash OR in-kind -> **sharecropping included**)
`parcel_rentedout`	`F01` land use	`== 6` ("rented out")
`parcel_purchased`	`F06` acquisition	`== 1` ("Purchased")
`parcel_certificate`	`F05` tenure status	per-wave title map (below); 1 titled / 0 untitled / . DK

Borrowed-in/out (`F01` 3 / 7) are non-market and are **not** counted as rented.

3. Parcel area (`parcel_area_ha`)

`hect` (area already converted to hectares); falls back to `F02` (amount) x `convert` (conversion factor; `F03` gives the unit - 1 lima, 2 acre, 3 hectare, 4 m²). Top-coded at `${area_max}` ha.

4. Design variables & identifiers

Variable	Source	Construction
`weight`	`field.dta` (2015/19) / `id.dta` (2012)	cross-sectional `weight/popwgt`; **2019 is panel-weighted** (see caveats)

Variable	Source	Construction
`ea_id`	`field.dta`	`cluster` (PSU)
`strataid`	`field.dta`	`group(dist)` (district stratum)
`parcel_id`	`field.dta`	`cluster-hh-field`
`year`	—	2012 / 2015 / 2019

5. `F05` tenure status - per-wave title map (the categories expand)

2012 has 6 categories; 2015 and 2019 share an expanded 9-category scheme, and the **code numbers do not line up** (2012 code 7 = "state land no title", but 2015/19 code 7 = "chief certificate"). The comparable **formal-title** flag is therefore built per wave:

	Titled = 1	Untitled = 0	DK / other / NC = .
2012	{1 state titled, 2 former-customary titled}	{3 customary no title, 7 state no title}	{4, 5}
2015 / 2019	{1, 2 state titled given/processing; 4, 5 former-customary titled given/processing}	{3 state not titled, 6 customary no title}	{8 DK, 9 other, -8 NC}

"Chief certificate" (2015/19 code 7) has no 2012 analogue. By **default it is NOT counted as a formal title** (set to 0) so the series stays comparable with 2012; set `global zmb_chief_cert 1` in `MASTER.do` to count it as a certificate instead.

6. Harmonization decisions & caveats (Zambia)

- **Field-level, single tier.** The field is the unit; `field.dta` is the whole-roster frame (owned, rented-in, borrowed-in, rented-out, borrowed-out, fallow, garden, …), so rates use an all-fields denominator and rented-out is directly observed.
- **Rented-in includes sharecropping** (the "rented in" land-use code is explicitly *cash or in-kind*); rented-in/out are read from land use `F01`, not the acquisition item.
- **Certificate categories expand 2012 -> 2015/19** and are aggregated to a comparable formal-title flag (table above); "chief certificate" excluded by default.
- **2019 weighting.** This release of RALS 2019 carries only a single `weight` labelled "Panel Weight" - there is no population/cross-sectional weight in the field/household/id files. 2019 estimates therefore represent the **followed panel** (with attrition), not a fresh cross-section; 2012 uses its `weight` and 2015 its `popwgt`. Treat the 2019 level with that caveat (point estimates only mildly affected; representativeness differs).
- **Deterministic area** via `hect` (already converted); top-coded at 40 ha.
- **Non-LSMS source.** RALS is fielded by IAPRI, not the World Bank; obtain it from IAPRI rather than the LSMS catalog. Validated by a Python replication of the extractor against the raw files (100% weight match; rates within expected ranges for all three waves).
